NUTIQ – AI Speech Analysis Module

Technical Implementation Report

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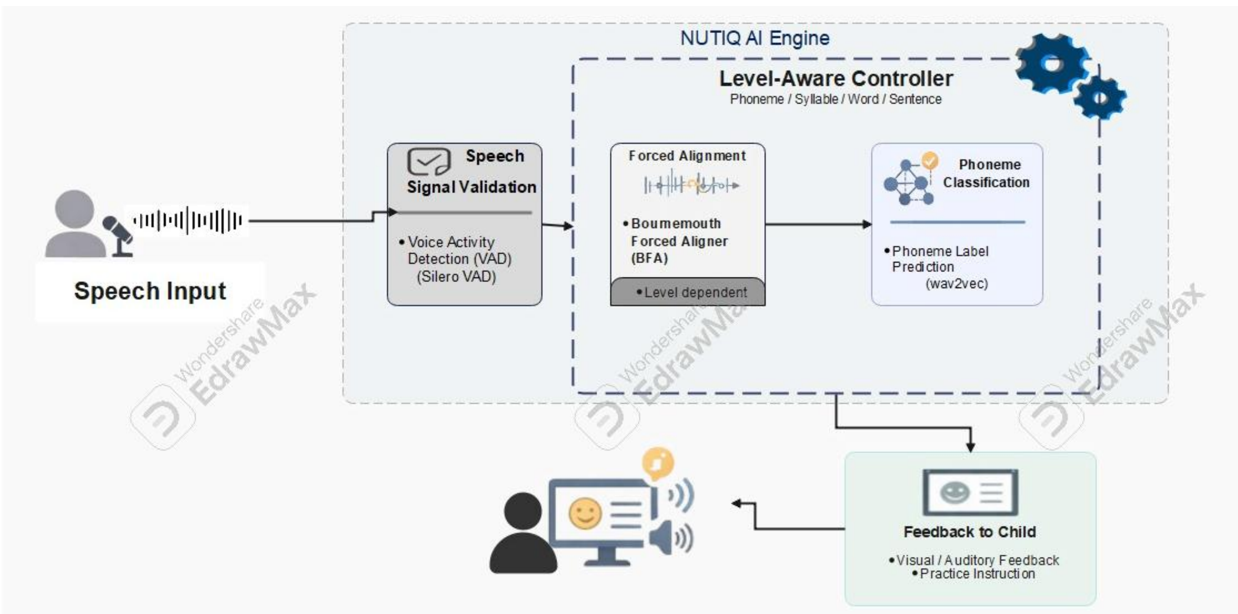
1. Introduction

The AI module in **NUTIQ** is designed to analyze human speech and detect pronunciation errors at multiple linguistic levels (phoneme, word, and sentence).
The system aims to simulate a real speech therapy assistant by providing accurate, real-time feedback based on advanced speech processing and deep learning techniques.



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2. Problem Statement

One of the primary challenges in speech-based systems is handling **raw audio recordings**, which often include:

- Silence segments
- Background noise
- Irrelevant audio artifacts

These factors negatively impact model performance and reduce classification accuracy.

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3. Preprocessing– Voice Activity Detection (Silero VAD)

To address this issue, a **Voice Activity Detection (VAD)** model was integrated.

Objective:

- Detect and extract only the segments containing speech
- Remove silence and irrelevant noise

Impact:

- Improved data quality
- Reduced model confusion
- Enhanced overall classification performance

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4. System Architecture

The system follows a **multi-level processing architecture**:

Level 1-2: Phoneme & Syllable Analysis

- Input: single phoneme or short sound
- Process: direct classification
- Output: predicted phoneme + accuracy score

Level 3-4: Word & Sentence Analysis

- Input: full words or sentences
- Process:
    - Alignment
    - Context-aware analysis
- Output:
    - Pronunciation correctness
    - Error localization

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5. Core Model – Speech Classification

A deep learning model based on **wav2vec 2.0** was used.

Why wav2vec 2.0?

- Handles raw audio effectively
- Learns rich speech representations
- Eliminates need for manual feature engineering
- Suitable for phoneme-level classification

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6. Dataset Preparation

The dataset was constructed from:

- Public datasets (e.g., Kaggle)
- Real-world recordings collected manually

Importance:

- Captures real pronunciation variability
- Improves model generalization

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7. Data Augmentation Strategy

To simulate real-world recording conditions, multiple augmentation techniques were applied.

Each audio sample was expanded into ~5 augmented versions.

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8. Real-World Simulation of Augmentation Techniques

Technique	Real-World Simulation
Gaussian Noise	Environmental noise (device noise)
Pitch Shift	Different voice tones (child, adult, gender variations)

Technique	Real-World Simulation
Time Stretch	Fast or slow speech patterns
Gain	Volume variation due to microphone distance
Shift	Delayed speech start / recording latency
Bandpass Filter	Low-quality microphones or phone calls
Background Noise	Real environments (home, café, street)
Clipping Distortion	Microphone overload / signal clipping

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9. Training Strategy

- Data augmentation applied dynamically
- Each augmented sample validated against original
- Training performed on balanced dataset

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10. Model Performance

Evaluation Results:

- **Test Accuracy:** 97.39%
- **F1 Score:** 97.38%
- **Validation Accuracy:** ~98.6%

Observations:

- High precision across most phonemes
- Slight confusion between phonetically similar sounds (expected behavior)

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11. Alignment Model (Advanced Processing)

A forced alignment model was integrated to support higher-level analysis.

Purpose:

- Detect the exact position of each phoneme within a word or sentence

Use Case:

- Identify:
  - Mispronounced phonemes
  - Position of the error within the word

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12. Intelligent System Logic

A decision-based logic system controls the pipeline:

Flow:

- If input = phoneme → Classification Model
- If input = word/sentence → Alignment + Analysis

Advantages:

- Efficient processing
- Dynamic model selection
- Scalable architecture

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13. Experiment Tracking

Training and evaluation were monitored using an experiment tracking system.

Capabilities:

- Track accuracy, loss, and F1-score
- Compare model versions
- Save best-performing checkpoints

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14. Conclusion

The AI module of NUTIQ successfully implements a robust speech analysis pipeline that:

- Cleans raw audio using VAD

- Classifies phonemes using a deep learning model
- Enhances robustness through realistic data augmentation
- Achieves high accuracy (~97%)
- Uses alignment techniques to detect pronunciation errors within context
- Adapts dynamically based on input type

This system forms a strong foundation for an intelligent, scalable speech therapy platform.

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15. References & Tools

- Silero VAD
  https://github.com/snakers4/silero-vad
- wav2vec 2.0
  https://huggingface.co/facebook/wav2vec2-base
- Audiomentations
  https://github.com/iver56/audiomentations
- Bournemouth Forced Aligner (BFA)
  https://github.com/tabahi/bournemouth-forced-aligner